Build-day checklist

Tick each box at the step it belongs to. Tape readings in cm, caliper and drill sizes in mm. Datum: top A's back-left corner (0, 0).

Measure before anything is cut, printed or drilled

- [ ] Both EKBACKEN tops 28 mm thick (calipers). If 27 mm, shorten every screw into the top by 1 mm. Never deeper than 21 mm into a top. **MEASURE ON HARDWARE**
- [ ] Feet pad diameter 50 mm (calipers) before printing the 52 mm bore rings. **MEASURE ON HARDWARE**
- [ ] ALEX height 70.0 cm (with and without its stock feet). ALEX bottom panel thickness, only if you fit the optional ALEX feet. **MEASURE ON HARDWARE**
- [ ] Delivered rail lengths (bought as 84 in, 213.4 cm; layout 213 cm) and angle thickness 4.8 mm. **MEASURE ON HARDWARE**
- [ ] VIVO lower clamp: maximum opening 31 mm or more (28 mm top + 3 mm plate); top arm reach (about 89 mm). **MEASURE ON HARDWARE**
- [ ] Room: 252 × 317 cm of floor cleared, including the 3 cm wall gaps; wall to wall 252.0 cm and 317.0 cm; baseboard thickness. **MEASURE ON HARDWARE**
- [ ] Feet load rating (design wants 50 kg or more each). **MEASURE ON HARDWARE**

Parts and tools on hand

- [ ] Six OLOV legs, 105 mm plate with five 5.5 mm holes on an 85 mm circle; 30 × M4 × 16 mm screws (H01), five per leg.
- [ ] Eight of the twelve 4 in feet: 59.5 mm plate with four 5.2 mm holes 46.8 mm apart; 32 × #8 × 5/8 in pan head (H02).
- [ ] Eight printed rings (52 mm bore, 3 mm wall, 10 mm ring on a 75 × 2.5 mm flange), PETG or ABS, flange-down; 24 × #6 × 5/8 in flat head (H06).
- [ ] Two rails, 2 × 2 × 3/16 in angle, ends deburred; 19 × #10 × 1 in pan head Robertson (H03).
- [ ] Four 10 × 3 in mending plates (two joining, two clamp pads); 8 × #10 × 5/8 in (H04) with #10 washers (H05).
- [ ] Two gusseted 90° strut fittings; 6 × 3/8-16 × 1-1/2 in grade 5 bolts (H08), 12 × 3/8 washers (H09), 6 nylon-insert nuts (H10).
- [ ] Printed tools: edge T-square, leg jig, feet corner jig, ALEX corner template, rail marking gauge, rail drill jig, two fitting centring plugs, drill guide (tower + 5 pucks), foot height gauge (102 mm), wall gauge (6.5 cm and 3 cm legs).
- [ ] Automatic centre punch, cutting oil; 5/64, 1/8, 7/32 and 7/16 in bits (7/16 cobalt); Robertson #2 bit on a 3 in extension.

Verify before: at each step

Step 1 · Steel on the bench

- [ ] Rail A holes at 3.5, 28, 54, 80, 106, 132, 158, 184, 210.5 cm from the wall end (first at 3.5, not 2.5); rail B at 2.5, 7.5, 28 … 210.5 cm from the corner end; all 19 mm from the flat leg's free edge.
- [ ] Fitting as template: rail A heel on the 52.0 cm mark legs toward the far end, and on 51.5 cm legs toward the wall end; rail B heel on the corner end. Holes 29 mm below the heel edge, 7/16 in.

Steps 2–3 · Marking the tops, face down

- [ ] Unfinished edge to the wall (y = 0 on top A, x = 0 on top B); the side-wall end of top A taped and measured from.
- [ ] A mark off by more than 2 mm: redraw it.

Step 4 · Legs, feet and rails

- [ ] Every OLOV pre-set to 80.2 cm, or the ALEX-feet table's number (decided before this step).
- [ ] Every foot set with the 102 mm gauge: slides over from the side with no rock.
- [ ] Rail A: 12 cm from the front edge to the vertical leg's outer face at both ends and in the middle before drilling; end screws first, re-check, then the rest. Rail pilots 25 mm from the flat leg's top face (20 mm into the wood).
- [ ] Rail B: holes 3 to 10 only; holes 1 and 2 wait for step 7.

Step 5 · ALEX units

- [ ] ALEX 1 back 6.5 cm off the back wall (wall gauge), near side 212.5 cm from the side wall; ALEX 2 side 6.5 cm off the side wall, near end 277.5 cm from the back wall.
- [ ] ALEX placed by the walls, no rings yet (the rings go on in steps 6–7, on the traced pad circles).

Steps 6–7 · Stand the tops, make the joint

- [ ] Each top stood on its bare ALEX, 3.5 cm of top showing all round (wall gauge's 6.5 cm leg at the wall); pencil traced round each pad; top lifted off.
- [ ] Each ring set on its circle, pencil line visible all round through the open bottom; pilots 3–4 mm through the ring's own holes, finished with the 12 ALEX puck; three #6 × 5/8 in screws.
- [ ] Top stood back up: all four pads drop into their rings. Top A 3 cm off both walls, level along x and y, all four pads bearing.
- [ ] Top B's legs 2–3 mm short before sliding rail B under top A; then wound up to flush.
- [ ] Before the plates: rail B's end touches rail A's vertical leg with no gap, and its outer face lines up with the 52.0 cm mark.
- [ ] Surfaces flush by the feet, never by packing under the joining plates.
- [ ] Before the bolts: surfaces flush and rail B touching rail A; inside fitting gusset-down. Six bolts 25 N·m.

Step 8 · Mount and trim

- [ ] Pad plates on their outlines (x 20.0–45.4 cm, y 0.5–8.1 cm; y 20.0–45.4 cm, x 0.5–8.1 cm); clamps finger-tight at x = 40.0 cm and y = 40.0 cm on Option 1 holes until the crossbar is on.
- [ ] G5 top edge 66.9 cm above the top, under the 67.3 cm limit; if the mount will not reach, lower the G5 (a bottom edge down to 12 cm above the top still clears the ELLOVEN).
- [ ] Trim: gap measured at five or six points per wall; gap set to the smallest minus 1 mm; desk slid back until the gauge's 6.5 cm leg just fits at each ALEX.

Step 9 · Final checks (chapter J)

- [ ] 1 · Surface 83.0 cm ± 0.3 cm at all six corners (85.0 / 87.0 / 89.0 cm with 2 / 4 / 6 cm ALEX feet).
- [ ] 2 · Level across the joint reads flat both ways; a fingertip finds no step.
- [ ] 3 · 20 kg (case of water + books) on top B's inside corner, 10 cm in from both edges: drop ≤ 2 mm on a straightedge from L1's line; returns, no creak, nothing moved. Never stand or sit on the desk.
- [ ] 4 · All eight pads bear: none spins free or rocks when turned by hand.
- [ ] 5 · Straightedge across the joint at the inside corner: step written down: _____ mm.
- [ ] 6 · Sideways push along each wall: no sway beyond a few mm, no top moving on its ALEX.
- [ ] 7 · No screw tip visible or felt on either laminate face.
- [ ] 8 · Knees clear; rail lower edge 75.1 cm from the floor, 12 cm back from the edge.
- [ ] 9 · Both ALEX fronts open fully without touching a foot or a leg.

Two weeks · date _____

- [ ] Retighten every screw.
- [ ] Straightedge at the joint: _____ mm. Settled? Retighten the six fitting bolts to 25 N·m (18 ft·lb).
- [ ] Each printed ring and its three screws: no crack (reprint in PETG if cracked), screws tight.

One month · date _____

- [ ] Straightedge at the joint: _____ mm. Settled? Retighten the six fitting bolts.